



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 1 Examinations 2024-2025

Course Instance Code(s) 4BCT1, 1EM1, 1OA1, 1CSD1, 1MA1, 2MF1, 4BS2

Exam(s) 4th B.Sc. Computer Science and Information Technology
Erasmus, Study Abroad, Bachelor of Science
M.Sc. in Computer Science (Data Analytics)
M.Sc. in Computer Science (Artificial Intelligence)
M.Sc. in Software Design and Development

Module Code(s) CT4100
Module(s) Information Retrieval

Paper No. 1

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Instructions: Answer any three questions. All questions carry equal marks

Duration 2 hours
No. of Pages 3
Discipline(s) Computer Science

Requirements:

Release in Exam Venue	No [☐]	Yes [☒]
MCQ Answer sheet	No [☒]	Yes [☐]
Handout	No [☒]	Yes [☐]
Formulae & Tables*	No [☒]	Yes [☐]
Cambridge Tables 2 nd Edition**	No [☒]	Yes [☐]
Graph Paper*** A4 Graph Paper 1mm 0.1cm Squared (Standard)	No [☒]	Yes [☐]
Other Materials	No [☒]	Yes [☐]
Graphic material in colour	No [☒]	Yes [☐]

PTO

Question 1

- (a) In relation to the vector space model, explain the purpose of normalisation. Outline any suitable approach to normalisation and discuss its effectiveness. (7)
- (b) One limitation of the vector space model is the assumption of term independence. Explain how this assumption may affect the model's ability to capture semantic relationships between terms. Illustrate this issue with a suitable example and outline an approach to address this limitation. (10)
- (c) With reference to a suitable set of axioms, explain how an axiomatic approach can be used to develop a good weighting scheme in information retrieval. (8)

Question 2

- (a) Explain, with examples, a technique that can be used to predict query difficulty using each of the following sources:
 - i) The query
 - ii) Collection wide evidence
 - iii) The returned ranked list of documents(9)
- (b) Explain briefly the Rocchio method in the context of relevance feedback. Discuss any limitations of this approach. (7)
- (c) Outline an approach to generating suggested terms to a user to augment their query. Assume there is no user feedback available, and that you wish to present a set of diverse terms to the user. (9)

Question 3

- (a) The extended Boolean model has been used to overcome some of the limitations of the classical Boolean model. Show how you would calculate the relevance of a document d to each of the following queries:
 - i) *street AND legal*
 - ii) *street OR legal*(10)
- (b) Given a ranked answer set and evaluation judgements, explain how you would generate a precision-recall graph. (5)
- (c) Learning mechanisms have been used successfully in information retrieval. Using an approach of your choice, suggest a learning mechanism to identify good weights to apply to terms in documents. Discuss any limitations of your approach. (10)

Question 4

- (a) Describe briefly an approach to collaborative filtering. Discuss the advantages and limitations of collaborative filtering techniques. (8)
- (b) Given a set of scientific articles (which contain title, abstract, authors, key words, main body of the paper, and a bibliography), suggest a suitable approach to measure the similarity between these documents in the collection. (9)
- (c) With reference to any two information retrieval tasks, explain how clustering algorithms can be of use. Your answer should include a description of the clustering approach used. (8)

THE END